

Adaptive Decentralized Emergent Behavior Based Autonomous Platooning

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Abstract—Vehicle platooning has been shown to significantly improve road safety and driver experience, as well as increase road capacity, reduce fuel consumption, and thus green house emission. We present the Adaptive Decentralized Emergent-based Platooning system, or ADEPT for short, an emergent behavior based autonomous platooning approach that adopts a bottom-up, biologically inspired approach where each vehicle follows a set of simple rules to execute platoon maneuvers. In this paper, we evaluate ADEPT's network performance against centralized platooning using a platooning simulator and show that ADEPT is able to yield superior network utilization. Additionally, we study the interplay between our emergent platooning system and vehicle characteristics and dynamics such as engine performance, weight, center of mass, friction, speed, and acceleration, and evaluate using a robot simulator equipped with an open dynamics physics engine. Experimental results demonstrate that ADEPT's controller is string stable when subject to different maneuvers and disturbances. In addition to evaluating ADEPT's controller stability, we propose methods that enable ADEPT to handle real-world conditions such as curved roads, vehicle following and obstacle avoidance. Our experimental results show that our dynamic radius of curvature determination method helps maintain inter-vehicle gap at 99% of the desired gap including on curved roads. Even during disturbance events, the gap errors are kept within the desired deviation. Using inspiration from lane assist technology, we propose a vehicle following technique that, according to our experimental results, yields high accuracy.

Index Terms—Autonomous-platooning, decentralized, platoon-maneuvers, bio-inspired, emergent, merge.

I. INTRODUCTION

THE Federal Highway Authority estimates that approximately 172 billion gallons of gas being consumed every year [1] releasing 3371 billion pounds of greenhouse gases into the atmosphere each year, mainly in the form of CO₂ [2]. Additionally, the National Highway Traffic Safety Administration reported 26K fatal crashes in 2021 of which 3522 were due to distracted drivers, an uptick of 12% from 2020 [3].

Vehicle Platooning has been identified as a promising solution to help reduce fuel emission and improve road safety. The U.S. Department of Transportation defines platooning as “A coordinated operation of two or more vehicles via cooperative adaptive cruise control”, and the Mid-American

Association of Transportation Officials defines platooning as “The practice of electronically coupling multiple vehicles using v2v communication in order to significantly shorten gaps between vehicles”. According to [4], platooning reduces fuel consumption by up to 5% for the first vehicle and up to 10% for the following vehicles due to reduction in air-drag. Autonomous platooning reduces driver fatigue, thereby improving the safety of our drivers. Additionally, reduction in the inter-vehicle gap can result in a 166% increase in the road capacity [5], [6].

Platooning approaches can be classified as “Centralized” or “Decentralized” depending on how platooning is coordinated. Centralized platooning is characterized by the presence of a leader and suffers from single point of failure and serialization. Longer platoons lead to more demanding communication requirements as the information transmitted by the leader has to travel longer distances to reach all the vehicles in the platoon. This leads to higher packet loss thereby restricting platoon length. These drawbacks can be mitigated by decentralized platooning, where there is no leader and vehicles exchange information only with their immediate neighbors. Decentralized platooning systems fall into two main categories, namely “Deliberate” and “Emergent”. Deliberate decentralized platooning approaches take a top-down approach where high-level objectives, i.e., maneuvers such as join, exit or merge are defined and then specific procedures and protocols to execute each maneuver are developed. On the other hand, emergent decentralized platooning takes a bottom-up approach, where platooning maneuvers emerge as platooning vehicles follow basic rules of interaction, termed “Emergent Rules”. This is called *Behavioral Shaping*.

Emergent platooning's salient characteristics include: (1) There is no leader, (2) All interactions called *Stigmergy* happen indirectly, and (3) Platooning vehicles react only to local information and none have a global view of the system. As discussed in more detail in Section III, these features result in important benefits including flexibility, reduced complexity and maneuver time.

In this paper, we present ADEPT, our Adaptive Decentralized Emergent-based Platooning, an emergent behavior based autonomous platooning system. ADEPT adopts a bottom-up, biologically inspired approach where each vehicle follows a set of simple rules in order to execute platoon maneuvers. This work builds on our prior work described in [7] and [8] and makes the following unique contributions:

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- 1) Defined the basic set of emergent rules sufficient to implement the three main freeway platooning maneuvers, namely join, exit and merge.
- 2) Developed and evaluated an efficient inter-vehicle gap calculation method suitable for curved roads which calculates the road's radius of curvature dynamically.
- 3) Developed and evaluated an efficient vehicle following model using camera images, inspired by the lane assist mechanism.
- 4) Developed an obstacle avoidance approach using LiDAR images.
- 5) Evaluated the interplay between emergent platooning and the underlying wireless communication network using a cooperative driving simulation environment.
- 6) Using a physics simulator, evaluated the interplay between emergent platooning and vehicles' dynamics considering different types of maneuvers and vehicles.

The remainder of this paper is organized as follows - In Section II, we discuss background and related work followed by a detailed description of ADEPT's design, components and features in Section III. Section IV describes ADEPT's emergent maneuvers. Performance evaluation results are discussed in Sections V and VI followed by conclusion in Section VII.

II. BACKGROUND AND RELATED WORK

Early research on platooning and autonomous vehicular systems mainly focused on steady-state cruising which controls the speed and inter-vehicular spacing. Projects like PATH [9], PROMETHEUS [10], SARTRE [11], Chauffeur [12] and KONVOI [13] all used static platoon creation, where members of the platoon are predetermined. A more dynamic approach aims to form the platoon at the freeway entrance using Road Side Units (RSUs) placed at the entrance of the freeways [14].

The problem of dynamically assigning vehicles to platoons as they drive on roadways, specifically on freeways, has been quite unexplored [15]. Only recently researchers have started to focus on platoon formation, maneuvering and dissolution. Dynamic autonomous platooning approaches can be classified into two main categories, namely centralized and decentralized. Below, we describe how these different approaches differ from each other and highlight some existing centralized and decentralized platooning systems.

A. Centralized Platooning

In centralized platooning systems, all platooning activities are coordinated by a *leader*, usually the vehicle at the front of the platoon. A notable example of centralized platooning is described in [16] which we use as a performance baseline in our experiments. Other examples include [17], [18], [19]. Platoons can also be formed with the assistance of infrastructure, e.g., RSUs [20], [21], [22]. Like any centralized system, centralized platooning suffers from drawbacks such as single point of failure and performance bottleneck (at the leader), as well as shorter platoon-length and data loss as the distance between the leader and the rest of the platoon increases due to various wireless communication impairments.

B. Decentralized Platooning

To mitigate the drawbacks of centralized platooning, in decentralized platooning systems, there is no platoon leader and vehicles interact only with neighboring vehicles to complete the maneuvers. Localized interactions allow decentralized platooning systems to handle longer platoons. Examples of decentralized platooning approaches include [15], [23], [24], [25].

Existing decentralized platooning systems adopt a “top-down” approach to platooning operations, i.e., formation, dissolution and maneuvering. This means that every operation and maneuver is handled by a procedure and protocol that are operation/maneuver specific. As a result, such “deliberate” decentralized platooning systems exhibit increased complexity, are harder to manage and extend.

C. Emergent Platooning

Emergent platooning is a decentralized platooning approach that aims at addressing the limitations of deliberate decentralized platooning systems. Like deliberate systems, in emergent platooning there is no leader, however, unlike deliberate platooning, emergent systems employ a “bottom-up” approach - they define and use a set of basic common rules that are used to implement all platooning operations and maneuvers. All participants take localized actions from which global behavior emerges. Emergent systems are inspired by biological system found in nature such as starlings, termites, ants, fish, etc.

Some early emergent system approaches include Craig Reynolds' pioneering work published in 1987 on simulating flocking birds, or *Boids*, by creating a virtual bubble around each Boid [26]. He used three basic rules

- 1) Repulsion to avoid collision of Boids that come too close.
- 2) Attraction to keep the Boids together.
- 3) Orientation to keep all Boids moving in same direction.

Roca et al. proposed a hierarchical version of emergent behavior to tackle the complexities of ultra-large scale systems (ULSS) by organizing ULSSs as a hierarchy of subsystems [27], [28], [29]. They use platooning as an example of ULSS and, in addition to the rules proposed by Reynolds [26], proposed the *Destination* rule used to group together vehicles headed to the same destination.

The system developed by Roca et al. targets unmarked free-ground where vehicles can move in an unconstrained manner and cannot be adopted in urban, suburban or other environments where vehicles need to drive on roadways. The main contribution of our work is to propose, develop, implement and evaluate an emergent behavior based autonomous platooning system suitable for roadway-based transit and transportation systems.

III. ADEPT OVERVIEW

In this section, we describe the design and implementation of a novel multi-vehicle decentralized platooning system named Adaptive Decentralized Emergent-behavior PlaTooning (ADEPT), which is inspired by the “emergent behavior” of

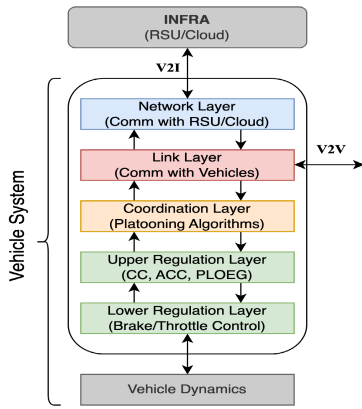


Fig. 1. Autonomous vehicle architecture [30].

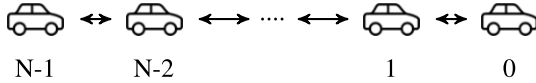


Fig. 2. Predecessor-Follower information flow model.

some processes found in nature. While in its current implementation, ADEPT focuses on autonomous platooning on freeways, highways and protected roads where maneuvers can happen during the trip, we plan to explore how ADEPT can be extended to operate in urban environments. ADEPT adopts the traditional autonomous vehicle architecture [30] shown in Figure 1. The Network layer is used to communicate with the underlying communication infrastructure (e.g., road-side units, freeway signs, etc.). ADEPT currently does not use any functionality in this layer. The Vehicle Dynamics layer represents the physical vehicle and its components such as throttle, brakes, wheels, etc., to which actuator inputs are provided by the Lower Regulation layer. ADEPT's emergent platooning features and functions are integrated into the Link layer, Coordination layer, and Regulation layer which are described in detail below. We assume that platooning vehicles are equipped with sensors like LiDAR, GPS, and camera and can communicate wirelessly directly with other vehicles using V2V communication.

A. Link Layer

The Link layer is responsible for the communication amongst the vehicles. ADEPT uses a decentralized *emergent behavior* based platooning approach where participants communicate indirectly via the environment mimicking emergent systems in nature, such as ants and termites. Therefore, in ADEPT, all communication is done via broadcasting in lieu of direct, unicast communication. ADEPT adopts the “Predecessor-Follower” information flow model illustrated in Figure 2. Platooning vehicles use predecessor information to adjust vehicle speed and when executing join and merge maneuvers. Follower information is used to decide when to become the tail-member and in inter-vehicle gap calculation. Vehicles in ADEPT communicate via *Stigmergy* using broadcast messages such as DSRC's Basic Safety Messages (BSMs) [31],¹ which are broadcasted by every participating

vehicle. The U.S. Department of Transportation recommends that BSM frequency should be at least 10Hz, which is adopted by ADEPT. According to the BSM specification [32], BSMs contain mandatory and optional fields. As shown in Figure 3, BSM's mandatory fields contain information such as vehicle ID, size, position, speed, acceleration, etc. Optional fields can be transmitted at the same or lower frequency. ADEPT uses BSM's optional fields to carry platooning information shown as grayed fields in Figure 3, adding 28 bytes to the BSM messages. Figure 3 also shows the fields that are used in platoon maintenance and fields that are used only during maneuvering. ADEPT's BSM messages are called *pheromone-messages*.

In ADEPT, to cope with the dynamics of platooning systems (e.g., high mobility, communication channel unreliability, etc.), vehicles only react after they receive a pheromone-message for a predetermined number of times. This is a configurable parameter which can be set based on factors such as the dynamics of the platooning system, the operational environment and the underlying communication infrastructure.

B. Coordination Layer

The Coordination layer is where ADEPT's platooning algorithms and platoon maintenance features are implemented. This layer is where most of our contributions reside. In ADEPT, platooning maneuvers emerge when vehicles follow ADEPT's emergent rules, which are described below.

1) *Emergent Rules*: Any emergent system consists of rules that are implemented by all participants, in our case, the vehicles. When these rules are executed by vehicles, a global pattern emerges. Here, we introduce the set of emergent rules that each vehicle in ADEPT follows to execute all platooning maneuvers. When a certain subset of these emergent rules is followed by each vehicle, the corresponding maneuver “emerges”, which is what characterizes emergent behavior based systems. Section IV describes how these rules are used to realize maneuvers in ADEPT.

The notation used in the description of the rules (see also Figures 8 and 9) is as follows: d is the minimum inter-vehicle gap, M_v is the merging-vehicle, S_{M_v} is the merging-vehicle's speed, S_p is the merged-platoon's speed, l_{M_v} is the length of M_v , G_{Fi} is the longitudinal-gap between M_v and F_i , and l_{Fi} is the length of vehicle F_i . Additionally, in ADEPT, at any given point in time, each participating vehicle has one of the following roles - NON-MEMBER: a vehicle that has not yet joined the platoon, JOINER: a vehicle that is in the process of joining the platoon, TAIL-MEMBER: the last vehicle of the platoon, and IN-MEMBER: vehicles currently in the platoon.

ADEPT's Emergent rules are as follows:

R1: Accelerate when the gap with the predecessor $> (d + \delta)$

R2: Decelerate when the gap with the predecessor $< (d + \delta)$

R3: Maintain the speed if the gap with the predecessor $= (d \pm \delta)$. This rule is also used by M_v to match its speed with merged-platoon by decelerating while $S_{M_v} > (S_p + \delta)$.

¹Even though we use the DSRC protocol in ADEPT's current implementation, ADEPT is agnostic of the underlying communication protocol used by vehicles.

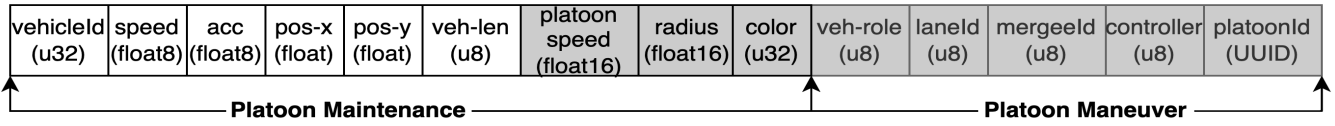


Fig. 3. ADEPT's beacon message.

R4: If no pheromone-message is received from the predecessor for n epochs, i.e., $n \cdot \delta t$ seconds, change controller to Cruise Control (CC) controller.

R5: If no pheromone-message is received from the successor for n epochs, i.e., $n \cdot \delta t$ seconds, become a TAIL-MEMBER

R6: Align steering angle with the visual angle of the predecessor.

R7: Used by M_v to determine F_i in the merged-platoon, i.e., the first vehicle that is longitudinally in front of M_v in the merged-platoon.

- If in-position then wait.
- If F_i is a one-vehicle platoon, brake until $G_{F_i} > d + l_{M_v}$ and re-initiate R7.
- If F_i is the first vehicle in multi-vehicle platoon, position behind the second vehicle of the merged-platoon and re-initiate R7.

R8: Used by F_i to create gap for M_v to merge by setting d , the inter-vehicle gap to be equal to $2d + l_{M_v} + G_{F_i} + l_{F_i}$, for as long as it senses its ID in M_v 's beacon. This gap is large enough for M_v with length l_{M_v} to merge safely.

R9: Rule is used by M_v to perform the lane change.

2) *Distance Calculation:* On a straight road, the gap between a vehicle and its predecessor can be calculated based on the euclidean distance between their positions. However, in real world, the roads are not always straight and have turns. Therefore, a more general and accurate approach to measure the inter-vehicle gap is needed. As previously noted, platooning vehicles are assumed to be equipped with GPS, which is a common feature in all newly manufactured vehicles. Consider a vehicle at position (x_2, y_2) and its predecessor at position (x_1, y_1) to be points on a circle with radius r as shown in Figure 4a. We observe that the euclidean distance d between the two vehicles will not accurately represent their inter-vehicle gap. However, actual distance between the two vehicles l is equal to $r\theta$, where θ is the angle subtended between (x_1, y_1) and (x_2, y_2) . The subtended angle θ can be calculated as $\theta = \cos^{-1}(1 - \frac{d^2}{r^2})$, where r is the radius of curvature of the road.

The next challenge is determining the radius of curvature (r). On a test track, r is known a priori. However, on regular roads, that is not the case. Therefore, in ADEPT, vehicles calculate r using the positions of both predecessor and successor as follows. Consider a vehicle, its successor and predecessor positioned on a circle forming the triangle whose sides have lengths a, b, c as shown in Figure 4b. When all three points of the triangle are circumscribed by a circle, then the circle's radius of curvature $r = abc/4A$, where A is the area of the triangle. A is calculated using Heron's formula; $A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$ is the

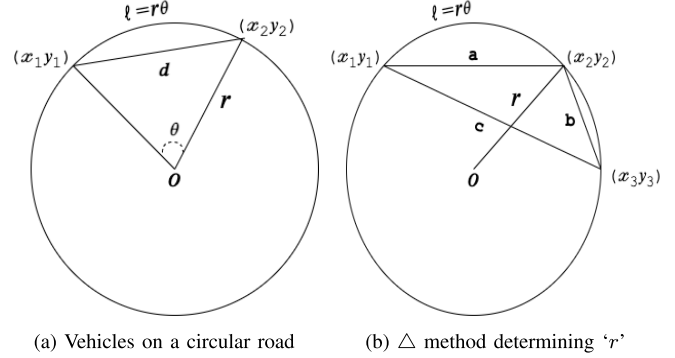


Fig. 4. Gap calculation.

semi-perimeter of the triangle. Note that the last vehicle of the platoon does not have a successor. Therefore, the radius of curvature calculated by each vehicle is sent in the pheromone-message and the last vehicle of the platoon uses the radius provided by its predecessor in its inter-vehicle gap calculation.

3) *Predecessor & Successor Determination:* Vehicles in ADEPT need to identify their predecessor and successor, e.g., to determine who is the source of received pheromone-messages. Additionally, when the vehicle exits the platoon, its successor should start using the information of the exiting vehicle's predecessor to determine the inter-vehicle gap and acceleration. Under challenging road conditions, e.g., when vehicles are driving on a winding road where direction is constantly changing, predecessor and successor determination becomes difficult.

ADEPT adopts the following simple approach for predecessor and successor determination that improves accuracy especially under challenging road conditions. Each vehicle keeps track of the positions it has visited. When it receives pheromone-messages from its neighbors, the position in the message is matched against the tracked positions visited by the vehicle. If a match is found, then the vehicle is identified as a successor, else it is considered a predecessor. This is true because only successors can see the positions the vehicle has visited. The number of positions (N) to track depends on various factors such as - (1) Number of successors to track (n), (2) Length of each vehicle (l), (3) Standstill distance between the vehicles (s), (4) Constant time headway of the platoon (h), (5) Platoon speed (v) and (6) Frequency of the pheromone-messages (δt). N is determined using the expression

$$N = \frac{n(l + s + h \cdot v)}{v \cdot \delta t} \quad (1)$$

Since a vehicle's tracked positions may not be exactly the same as its successor's, ADEPT's predecessor and successor determination algorithm considers a match if the position transmitted by the vehicle is within a small radius of the tracked positions. Once a vehicle categorizes its neighbors

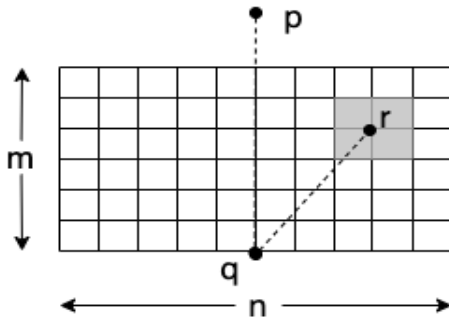


Fig. 5. Camera image representation.

as predecessor and successor, they are sorted based on their approximate distance (d) from the vehicle, where d is the average of the Euclidean and Manhattan distances.

4) *Vehicle Following*: In a platoon, except for the first vehicle, all other vehicles have to follow one another. While there are a number of ways to perform vehicle following, in ADEPT, vehicles determine their steering angle using images from their front cameras. Even though this method is not new, we provide a brief description of our approach here to highlight the different challenges posed by decentralized platooning systems. We assume that every vehicle is equipped with a camera whose refresh rate (δt) is higher or equal to the pheromone-message frequency. Every vehicle transmits its color in RGB format every δt seconds. The vehicle's controller uses the predecessor vehicle's color from the camera image to determine the steering angle. Note that, this method needs to be adjusted in real world scenarios to be able to differentiate black predecessors from the asphalt. For example, a high resolution camera that can process license plate information could be used. Additionally, if the license plate information is exchanged between the vehicles, it could also serve to verify that received messages are coming from the predecessor. ADEPT's vehicle following approach works as follows.

Consider the image shown in Figure 5, whose resolution is $m \times n$ pixels. Each block corresponds to a pixel and the gray rectangle represents the cluster of pixels corresponding to the predecessor vehicle as seen by the camera. To determine the steering angle, the vehicle's width is used (not its height), because it is assumed that vehicles are on the same plane. Therefore, only the horizontal distance from the center of the image need to be considered. Let P be the set of pixels in the cluster and $p_i \in P$ the pixels in that cluster. The steering angle (A) is given by -

$$A = \left(\frac{\sum (p_i \bmod n)}{|P| \times n} - 0.5 \right) \times FoV \text{ radians} \quad (2)$$

where $FoV = \pi$ is the camera's field-of-view. Negative angles imply leaning left, while positive angles represent lean right and zero implies going straight. In Figure 5, $n = 10$ and if $P = \{17, 18, 27, 28\}$, using Equation 2, we get $A = 0.78$ radians, i.e., turning right by 0.78 radians. Similarly, if $P = \{22, 23, 32, 33\}$, then we get $A = -0.78$ radians, i.e., turning left by 0.78 radians. To smoothen the steering rotation, a proportional integral derivative (PID) controller is used.

An interesting question is why the position of a vehicle is not sufficient to establish its steering angle. As illustrated

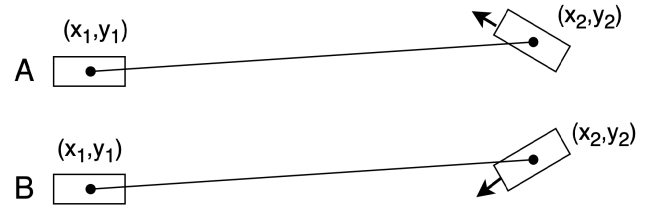


Fig. 6. ADEPT's steering angle calculation.

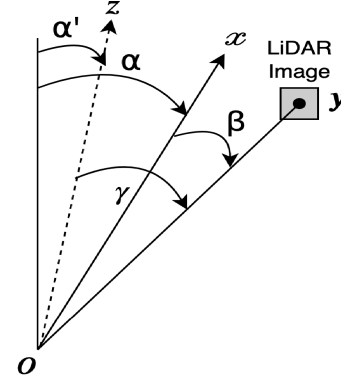


Fig. 7. Determining the steering angle for obstacle avoidance.

in Figure 6, the steering angle is based on the direction the vehicle is facing. In Figure 6's example, in both cases, the successor is in the same position (x_2, y_2) . However, in case (A) the successor has to turn left while in case (B) the successor has to turn right.

5) *Obstacle Avoidance*: ADEPT's current obstacle avoidance mechanism assumes that the first vehicle is equipped with a LiDAR camera.² The LiDAR captures range image containing the distance of each pixel in the image. Only the center portion of the image is used to avoid non-interfering objects (e.g., guard rails, trees or buildings) to be considered. Additionally, only objects up to m meters are considered. The obstacle's angle is calculated using the same method as the one described in Section III-B4.

Referring to Figure 7, let $\vec{ox}$ represent the direction in which the vehicle is moving with current steering angle α . The gray box represents the obstacle and its angle with respect to the vehicle is β . Let γ be the minimum required clearance angle from the obstacle. Therefore, the vehicle has to steer and start moving in the direction of $\vec{oz}$ at an angle α' to ensure clearance from the object. To ensure smooth turning, the steering angle is made inversely proportional to the obstacle distance, i.e. $|oy| = d$. Thus, the new steering angle α' is given by

$$\alpha' = \alpha + (\beta - \gamma)/d \quad (3)$$

Similarly, if the obstacle is to the left of $\vec{ox}$, the new steering angle α' is given by

$$\alpha' = \alpha + (\beta + \gamma)/d \quad (4)$$

²Since vehicles without LiDAR may also join the platoon, they would not be allowed to be the platoon's first vehicle. Alternatively, they would not be allowed to join or have a rule that if a vehicle without LiDAR becomes the first vehicle after all its predecessors exited the platoon, it should also exit the platoon and can rejoin as long as it is not the first vehicle.

C. Regulation Layer

The Regulation layer consists of two sub-layers, namely the upper and lower regulation layers as shown in Figure 1. Information such as acceleration and position received from the neighboring vehicles via the Link layer is fed to the upper regulation layer. This layer determines the vehicle's acceleration and position, that in-turn is used to maintain the required inter-vehicular gap. The acceleration or deceleration determined by the upper regulation layer is passed to the lower regulation layer which then determines the pressure to be applied to brakes or throttle.

1) *Controllers*: The controllers used in the upper regulation layers determine the vehicle's rate of acceleration/deceleration. The first vehicle of the platoon does not have a predecessor and uses its Cruise Control (CC) controller, a simple proportional controller, to maintain the desired speed [30]. The acceleration ($\ddot{x}_i$) is determined using the equation:

$$\ddot{x}_i = -k_d \cdot (\dot{x}_i - \dot{x}_{des}) \quad (5)$$

where $\dot{x}_{des}$ is the desired platoon speed, $\dot{x}_i$ is the current speed of the vehicle and k_d is the proportionality constant that determines the rate of acceleration or deceleration.

In ADEPT, the other vehicles in the platoon determine their speed using a decentralized controller approach proposed by Ploeg et al. [33]. Alternatively, other controllers such as the one proposed by Liu et al. [34], could be used as well. The Ploeg controller determines the rate of change of desired acceleration ($\dot{u}_i$) using the equation

$$\dot{u}_i = \frac{1}{h} (-C_1 \cdot u_i + k_p \cdot e_i + k_d \cdot \dot{e}_i + C_2 \cdot u_{i-1}) \quad (6)$$

where h is the platoon's constant headway time, u_i the vehicle's desired acceleration determined by the controller, e_i the gap error. The latter is defined as the difference between the platoon's desired inter-vehicle gap and the actual gap with the vehicle's predecessor determined using the approach explained in Section III-B2. $\dot{e}_i$ is the speed error which is the difference between the ego vehicle's (vehicle of interest) speed determined using onboard sensors, and the predecessor's speed received via wireless communication. u_{i-1} is the predecessor's desired acceleration received via wireless communication. The vehicle's desired acceleration is then determined by multiplying the rate of change of desired acceleration by the time interval (δt) between two epochs, i.e., $u_i = \delta t \cdot \dot{u}_i$. C_1, k_p, k_d, C_2 are tunable constants explained in more detail in Section V-C under 'Hyper-Parameter Tuning'.

ADEPT's current platooning maneuvers include **Join**, **Exit** and **Merge**. In this section, we describe these maneuvers in detail and how they are realized using ADEPT's Emergent rules introduced in Section III-B1.

D. Emergent Join

During ADEPT's join maneuver, the joining vehicle J_v and the platoon are on the same lane as illustrated in Figure 8. Joining vehicles are by default in the TAIL-MEMBER role and self-assign a globally unique platoon-id, e.g., a UUID [35]. When a vehicle enters a

freeway, it uses the CC controller and scouts for platoons ahead. The speed of the single-vehicle platoon is set to $RANDOM(MaxPlatoonSpeed, MaxFreewaySpeed)$, where $MaxPlatoonSpeed < MaxFreewaySpeed$. This allows the single-vehicle platoon to catch up to a target multi-vehicle or single-vehicle platoon to have an opportunity to join. When J_v hears pheromone-messages from the TAIL-MEMBER of the target platoon (F_{i+1}) and is within the platooning distance from F_{i+1} , J_v sets F_{i+1} 's *vehicleId* as the *mergeId* in its pheromone-message, transitions to JOINING state and sets its role to JOINER. The vehicle's role is indicated in the pheromone-message.

On noticing its *vehicleId* in the received pheromone-message from J_v , F_{i+1} sets J_v 's *vehicleID* as *mergeID* in its own pheromone message and transitions to the TRANSIT state. This process continues until sufficient pheromone-messages are gathered with *mergeId* set to its *vehicleId*. Once sufficient pheromone-messages are gathered, both F_{i+1} and J_v transition to FINALIZING state. In this state, J_v changes its role to TAIL-MEMBER (F_{i+2}) and adopts the *platoonId* of the platoon. Vehicle F_{i+1} changes its role to IN-MEMBER.

ADEPT's Join uses Emergent Rules R1, R2 and R3 as follows:

- The new TAIL-MEMBER (F_{i+2}) notices that the gap with its predecessor (F_{i+1}) is greater than the platoon's inter-vehicle gap (d). It accelerates to catch up with the predecessor due to rule **R1**.
- Once it reaches the desired gap, F_{i+2} maintains the desired speed using rules **R1, R2, R3**.
- All vehicles, except the platoon's first member, maintain their speed using rule **R3**.

E. Emergent Exit

Any vehicle can leave the platoon without taking any specific action (e.g., inform other vehicles). When exiting the platoon, the vehicle generates a new globally unique *platoon-id* which will be broadcasted as part of the vehicles' new pheromones (i.e., once it leaves the platoon). Pheromones received with different *platoon-ids* are dropped by the other vehicles. Platoon illustrated in Figure 8b is reused to explain ADEPT's Emergent exit maneuver below.

ADEPT's Exit uses Emergent Rules R1, R4 and R5 as follows:

- When the TAIL-MEMBER (F_{i+2}) leaves the platoon, F_{i+1} will not receive any pheromone-messages from its successor. Therefore, successor pheromone-message concentration becomes zero and F_{i+1} becomes the new TAIL-MEMBER according to rule **R5**.
- When an IN-MEMBER (F_i) leaves the platoon, F_{i+1} will no longer receive periodic pheromone-messages from F_i and predecessor pheromone-message concentration drops to zero. F_{i+1} will accelerate to catch-up with F_{i-1} according to rule **R1**.
- When the IN-MEMBER is F_1 , predecessor pheromone-message concentration in F_2 remain zero for an extended period. Therefore, F_2 will become the platoon's first vehicle using rule **R4**. It changes its controller to CC and

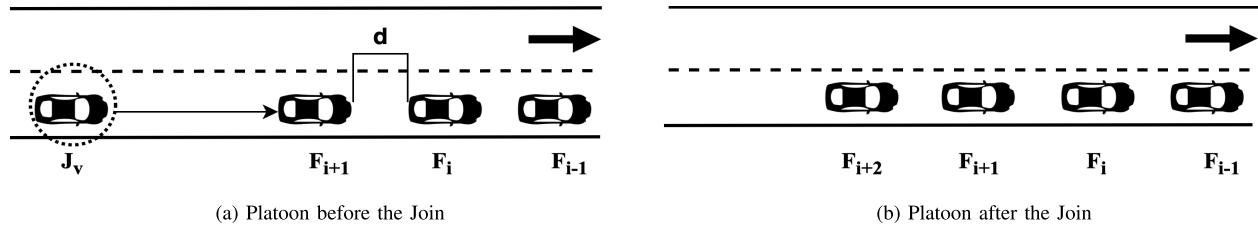


Fig. 8. Join maneuver.

maintains its speed independently without cooperation from other vehicles.

F. Emergent Merge

Consider two platoons – The one from which vehicles intend to leave is called the merging-platoon and the platoon to which vehicles intend to merge is called the merged-platoon. One of the main steps during a merge maneuver is identifying the target vehicle in the merged-platoon. As illustrated in Figure 9a, if the merging-vehicle M_v wants to merge, the target vehicle for the merge is the closest vehicle ahead of M_v in the merged-platoon. F_i will be the new following-vehicle after the merge. F_i creates a gap with F_{i-1} (as explained later in this section) so that M_v merges between F_i and F_{i-1} as shown in Figure 9b. Note that the closest rear vehicle F_{i+1} in the merged-platoon is not chosen because the position of M_v relative to F_i may be such that merging is not possible between F_{i+1} and F_i , even after F_{i+1} creates a gap with F_i as depicted in Figure 9c.

Alternatively, we could decelerate and reposition M_v for it to merge between F_{i+1} and F_i . This approach is more complex compared to decelerating only F_i as the former involves coordination among two vehicles while the latter involves only one. We reposition M_v when either F_{i-1} or both F_i and F_{i-1} are not present in the system. In the former case, i.e., the closest vehicle in the target platoon is the lead vehicle, M_v re-positions itself behind F_{i+1} because F_i will be the first vehicle i.e., F_0 , and it cannot create any gap with the vehicle in front of it. In the latter case, i.e., M_v is in front of the first vehicle in the target platoon, M_v will reposition itself behind F_{i+2} . In both cases, after repositioning, M_v will attempt to re-identify its new target vehicle. The new target vehicles will be F_{i+1} and F_{i+2} , in the former and latter cases, respectively.

If the merged-platoon is a single-vehicle platoon, F_0 cannot create a gap in front of it for M_v to merge, nor M_v can reposition behind a vehicle that can create a gap for it to merge. In this case, M_v positions itself at a safe distance behind F_0 and merges into the merged-platoon. We do not accelerate M_v as it may be not possible to do so because of the existence of a vehicle in front of M_v in the merging-platoon. Though it is possible to accelerate M_v if it is the first vehicle in the merging-platoon, we chose not to do so for implementation simplicity.

ADEPT's Merge uses Emergent Rules R1, R2, R3, R7, R8 and R9 as follows:

- In the IDLE state, all platoon vehicles maintain a minimum inter-vehicle gap of d using **R1**, **R2**, **R3**. This

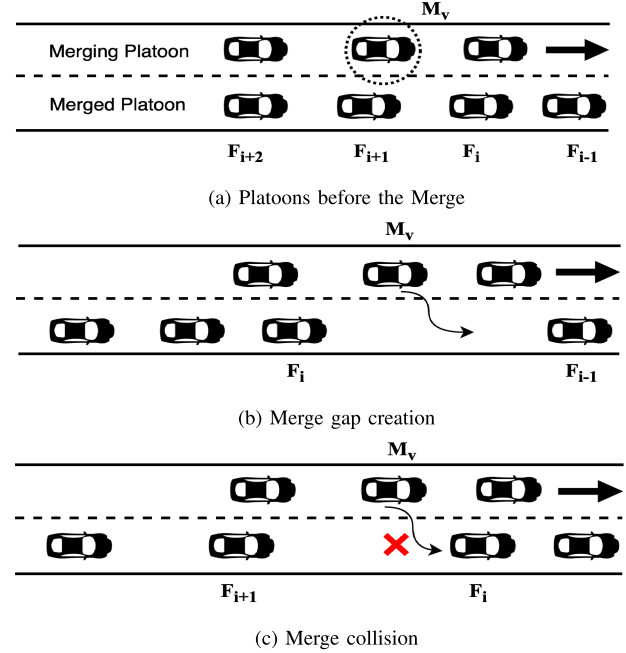


Fig. 9. Merge maneuver.

is achieved by the vehicle's controller, in our current implementation, the Ploeg controller.

- When M_v is ready to carry out the merge, it moves to the SPEED-MATCH state to match the speed of the target platoon using **R3**. The speed of the target platoon is extracted from the beacon message from one of the merged-platoon members.
- When M_v is in sync with the speed of the merged-platoon, it transitions to the FIND-TARGET state and uses **R7** to identify F_i , the nearest vehicle in the merged-platoon ahead of it. The beacon message from F_i contains the required length and position information. M_v identifies F_i by maintaining a sorted list of positions of all the nearby merged-platoon vehicles.
- Once M_v identifies F_i as the target vehicle, its own ID (mergeeId) is set in its beacon message and M_v moves to the GAP-WAIT state where it constantly checks the position of F_i , which is advertised in F_i 's beacon message.
- When F_i senses its ID in M_v 's beacon message, it moves to the GAP-CREATE state and uses **R8** to create the required gap with F_{i-1} . As specified in **R8**, the gap is set as the minimum inter-vehicle gap whereby the vehicle's Ploeg controller slows down the vehicle until the appropriate gap with F_{i-1} is created. Once the desired merge gap is created, F_i moves to MERGE-WAIT state.

In this state, F_i processes beacons only from M_v and maintains the new gap while its ID is present in M_v 's beacon message.

- M_v constantly monitors its environment by listening to the beacon messages sent by the nearby vehicles. Distance between M_v and the other vehicles is determined using the position information it receives in the beacon message from other vehicles. The merge is deemed safe when M_v does not sense any other vehicle within the merging distance in the merged-platoon for a pre-determined period of time. The merging distance is a configurable parameter which is set to $2l_{Mv}$. The controller capable of guiding the vehicle from one lane to another is used to complete the merge maneuver.
- M_v stops specifying F_i 's ID in its beacon message and moves to the IDLE state either when it successfully completes the merge maneuver using **R9** or when it times out waiting for safe condition to merge. In the latter case, the merge operation is aborted and may be retried by M_v later.
- When F_i no longer senses its ID in M_v 's beacon message, it transitions to the IDLE state and re-positions according to the original inter-vehicle gap. All vehicles in IDLE state maintain a minimum inter-vehicle gap with their preceding vehicle using rules **R1**, **R2**, **R3**.

IV. EVALUATING ADEPT USING A COOPERATIVE DRIVING SIMULATOR

In our previous work [7], we evaluated our novel emergent platooning system using Plexe [16], a cooperative driving simulation environment. It uses SUMO [36] and Veins [37] to simulate vehicular mobility and vehicular networks/communication, respectively. Veins implement the IEEE 802.11p protocol which is the basis for the DSRC protocol used in vehicular communication. SUMO implements several cooperative driving controllers used by autonomous vehicles. As discussed in Section III-C1, ADEPT uses the Ploeg decentralized cooperative driving controller. In [7], we demonstrated experimentally Ploeg's string stability and compared our emergent decentralized Join and Exit maneuvers against their centralized counterparts. We showed that Emergent join takes less time to execute compared to Centralized join. As the number of exiting vehicles increases in scenarios where multiple vehicles exit the platoon, the time to execute the Centralized exit also increases since the maneuvers are serialized. However, Emergent exits are independent of one another as exiting vehicles need not coordinate with others.

In [8], we implemented ADEPT's multi-vehicle multi-merge maneuvers and evaluated its performance in terms of maneuver time and communication overhead for six different merge scenarios. Our results indicate that ADEPT yields lower overall maneuver latency (up to 300% in the experiments we ran) for scenarios involving simultaneous multi-vehicle multi-manuevers. We also show that ADEPT's communication overhead is significantly lower in terms of the amount of data transmitted when compared to centralized merge (up to 76% for some scenarios) and thus is able to use network resources more efficiently.

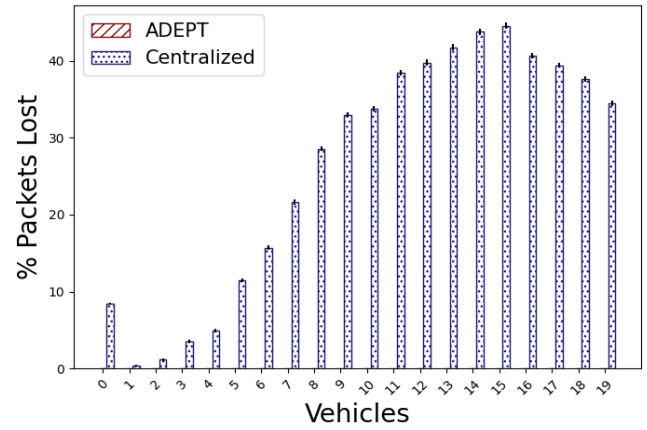


Fig. 10. Message loss ratio.

In this paper, we extend our evaluation of ADEPT's emergent platooning performance and focus on how efficiently ADEPT utilizes the underlying wireless communication network compared to centralized platooning. To this end, we conducted simulation experiments with a platoon of 20 vehicles. Once a simulation run starts, every 2s a new vehicle enters the scene until a total of 20 vehicles are present. The vehicles attempt to join the platoon as they enter the system. Details of ADEPT's platoon formation can be found in [8]. We collected data from 250 runs using different random seeds. To form the 20-vehicle platoon, it took an average of 96.8s in the centralized platooning approach and 89.7s in ADEPT. The statistics were gathered for a total of 300s of simulation time, which included platoon formation. To eliminate outliers, values within one standard deviation from the mean are considered. Error bars in the graphs represent the measurement sample's standard error, calculated as $SD/\sqrt{n}$, where SD is the sample's standard deviation and n is the number of samples, i.e., simulation runs.

A. Network Performance

In centralized platooning the lead vehicle has to communicate with all vehicles in the platoon, including the tail member, vehicles communicate with maximum transmission power (100mW in our experimental setup). In ADEPTs, however, vehicles only need to receive information from two of its immediate neighbors, which allows them to communicate at much lower power (0.3mW in our experiments).

1) *Message Loss*: Recall that in centralized platooning, in addition to beacon messages broadcasted periodically (in our current implementation, every 100ms), vehicles must also send unicast messages to the lead vehicle when performing maneuvers. In the case of the 20-vehicle platooning experiment we conducted, as vehicles enter the system and try to join the platoon, they unicast join request messages continuously to the leader until they are allowed to join the platoon. The graph in Figure 10 plots the message loss ratio for each platooning vehicle i calculated as $Loss_i = (MU_i - MR_i)/MU_i$, where MU_i is the total number of unicast messages sent to vehicle i and MR_i is the total number of unicast messages received by i . Since all vehicles in the platoon send unicast messages to

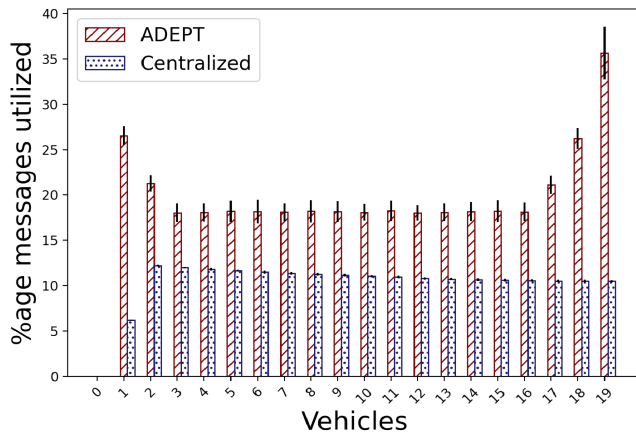


Fig. 11. Network utilization.

the lead vehicle, v_0 , the number of unicast messages v_0 lost is higher compared to all the other vehicles. However, as seen in Figure 10, v_0 's message loss ratio is lower compared to most of the other vehicles since the total number of unicast messages sent to v_0 is high. It is also noteworthy that the time vehicles take to perform the join maneuver and enter the platoon increases as number of vehicles in the system increases due to serialization at the leader. This results in more retries and higher message loss for the vehicles entering the system late as illustrated in Figure 10. However, the last few vehicles have to wait for less time since many of its predecessors would have already joined the platoon by the time they enter the system, thereby leading to somewhat lower message loss ratio towards the tail of the platoon. In the case of ADEPT, platooning vehicles do not need to transmit any unicast messages and therefore do not succumb to these losses. This is why ADEPT's message loss ratio shown in Figure 10 is zero for all vehicles in the platoon. As a consequence of not needing to use unicast messages, ADEPT generates less network traffic load and thus requires less resources from the underlying communication infrastructure.

2) *Network Utilization*: It is also important to characterize how much of the traffic load generated by the platooning system is actually consumed by the vehicles. Every vehicle transmits their position and acceleration information in the periodic beacon messages. In centralized platooning, each vehicle uses information from the leader and its predecessor to determine its own acceleration. However, since vehicles transmit at higher power, beacon messages end up being received by vehicles that do not necessarily need this information. As a result, most messages received are just dropped. In these experiments, on average, 10% of the messages across all vehicles that are received are utilized.

In emergent platooning, on the other hand, vehicles transmit beacon messages at lower power as only messages from their immediate predecessor are required to determine their own acceleration. Messages from other neighboring vehicles (e.g., from the second predecessor and a couple of successors in our experiments) are still received and then dropped. However, as shown in Figure 11, which plots the average percentage of messages utilized by each vehicle for both centralized platooning and ADEPT, the percentage of utilized messages

in ADEPT is much higher on average when compared to centralized platooning. In our experiments, ADEPT's message utilization is, on average, two times higher than that of centralized platooning.

Note that v_0 's network utilization is not shown as it does not determine its acceleration using information from other vehicles. Another interesting observation from Figure 11 is that in the case of centralized platooning, since v_1 immediate predecessor is the leader itself, v_1 only uses information from the leader but receives information from most vehicles. Therefore, v_1 's network utilization is lower when compared to the other vehicles, all of which show similar utilization.

In ADEPT, vehicles in the middle of the platoon receive beacon messages from the same number of vehicles and utilize information from only their predecessor. As a result, they all have similar utilization. However, vehicles at both ends of the platoon either do not have predecessors or do not have successors and therefore, show better utilization as they receive less number of unnecessary messages.

V. EVALUATING ADEPT USING A PHYSICS SIMULATOR

It is important to understand the interplay between ADEPT's emergent platooning and the physics of the underlying vehicle. In other words, the goal is to study the effect of the vehicle's physical characteristics such as acceleration, weight, center of mass, friction, etc., on the platooning system and algorithms including ADEPT's ability to handle real-world requirements such as driving on curved roads, vehicle following and obstacle avoidance.

For this study we considered two robot simulation platforms, viz., Gazebo [38] and Webots [39]. We chose the latter due to its ease of use and good documentation. Webots is an open source simulation software that enables simulation of robotic systems and provides complete development environment to model and program the robots. In the specific case of vehicular systems, Webots implements models for different vehicles, including but not limited to Tesla Model3, Range Rover, Lincoln MKZ, etc. It also has built-in rudimentary wireless communication support which enables vehicles to tune to a channel, broadcast and receive messages on that channel. However, simulating autonomous vehicle platooning where platoon members as well as other vehicles cooperate by exchanging messages require more complex abstraction of the underlying communication network. Network simulators such as OMNeT++ [40] and ns-3 [41] provide adequate modeling of the network. Implementing Webot functionality and features "natively" in an existing network simulator or implementing a network simulator inside Webots, would be both labor- and time-intensive. As such, we used our Network Simulation Bridge, or NSB [42] to bridge Webots with the OMNeT++ network simulator. NSB is a simple, low-overhead pipeline consisting of a message server and client interface libraries that bridge together applications and network simulators. NSB is platform-agnostic and uses a "plug-and-play" approach which allows it to integrate any application front-end with any network simulator back-end. An overview of NSB is provided in Section V-A. Section V-B summarizes the general setup

used in our experiments. Sections V-C, V-D, V-D2, and V-E present our experiments and results of our performance evaluation of ADPET's controller, ADEPT's ability to drive on curved road environments, ADEPT's predecessor following feature and ADEPT's maneuvers, respectively.

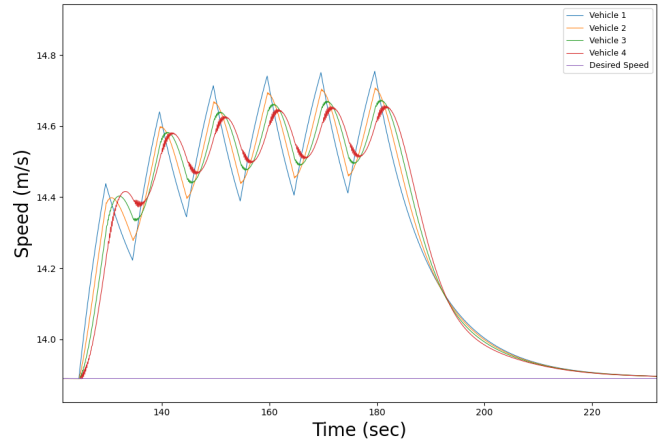
A. NSB

The Network Simulator Bridge – NSB consists of three main components, namely, the *NSB Server*, the *NSB Application Client* and the *NSB Simulator Client*. The NSB Application Client is used to interface the NSB Server with the front-end application, in this case Webots, while the NSB Simulator Client interfaces with the back-end network simulator, which in our current implementation is OMNet++. In our Webots implementation of ADEPT's platooning, participating vehicles communicate with one another via the underlying communication network simulated by the network simulator back-end, i.e., OMNet++. This communication is facilitated by the NSB-Server which functions as a messaging server. Therefore, for every vehicle $\{V_0, V_1, \dots, V_n\}$ in Webots, a corresponding wireless node $\{N_0, N_1, \dots, N_n\}$ is instantiated in the network simulator. For every such node pair (V_i, N_i) , the NSB Server maintains a set of three queues, viz., (1) Transmit Queue (TX-Q), (2) Transit Queue (TR-Q) and (3) Receive Queue (RX-Q).

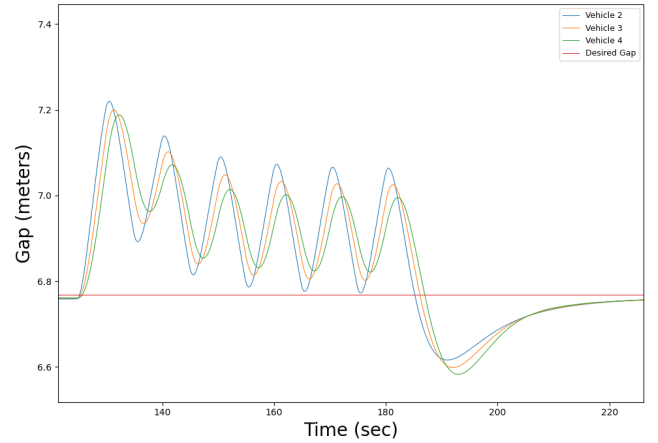
At the start of a simulation run, the NSB Application Client and the NSB Simulator Client register their nodes with the NSB Server. Then, the NSB Server associates the application with the simulator nodes in the order they were registered and maintains a table that maps application node IDs (e.g., IP addresses) to simulator node IDs (e.g., UINT32). The application and simulator libraries provide the necessary functions to Webots and OMNet++ to communicate with the NSB Server. Messages sent by a Webot node (V_i) are queued in its TX-Q until they are retrieved by the corresponding network simulator node (N_i). Before being added to the node's TX-Q, every message is assigned a 32-bit unique MsgID that monotonically increases. N_i retrieves messages by polling the TX-Q and messages are then moved to the TR-Q. Messages are routed to the receiver node N_j through the simulated network and are then enqueued in the RX-Q of the corresponding application node V_j . V_j polls the NSB Server for the messages from N_j . Messages are discarded by the NSB Server after being delivered to V_j . Complete details of the NSB architecture, protocol and messages are described in [42].

B. Experimental Setup

In all our experiments, each vehicle is fitted with a GPS sensor to determine the vehicle's position, a LiDAR camera for obstacle detection and a video camera for predecessor following. We experimented with different vehicle models to study how different vehicles' physics affect ADEPT's performance. We considered the Tesla Model3, the Range Rover and the Lincoln MKZ models. Experiments were conducted using Webot R2022b integrated with the OMNeT++ 5.6.2 network simulator via NSB 1.0 (as described in Section V-A).



(a) Vehicle's response to disturbance: speed adjustment



(b) Vehicle's response to disturbance: inter-vehicle gap adjustment

Fig. 12. Platoon response to disturbance.

C. Controller Performance

In our experiments, the platoon's desired speed v is set to 50kmph (13.88m/s). The horizontal line in Figure 12a represents the desired speed of 13.88m/s. The desired inter-vehicle gap is a function of desired speed, desired time spacing and an absolute minimum distance between the vehicles. The minimum distance r is set to 4.0m, which corresponds to the average length of the vehicles used in the experiments. Desired time spacing of $h = 0.2s$ was chosen after experimenting with a wide range of values as it is the minimum spacing required for platoons to be stable and not to crash. Given the above values, the desired inter-vehicle gap is calculated as $r + h \cdot v = 6.78m$, which is a constant for homogeneous platoons and is shown as a horizontal straight line in Figure 12b.

One of ADEPT's main components is the Ploeg controller represented using the Equation 6, whose performance is determined by analyzing the overall platoon stability, in particular *individual vehicle stability* and *string stability*.

1) *Individual Vehicle Stability*: A platoon system is said to provide "individual vehicle stability" if, when the lead vehicle driving at constant speed/velocity, i.e., its acceleration $a_0 = 0$, all $(n - 1)$ vehicles in the platoon can maintain the desired inter-vehicle gap. This can be observed in Figures 12a and 12b prior to 120s and post 240s.

TABLE I
CONTROLLER PARAMETERS

	Parameter	Value
CC	k_p	0.1
	Max Acceleration	$1.5m/s^2$
	Max Deceleration	$3.5m/s^2$
Ploeg	Platoon size	4 vehicles
	Platoon speed	50km/h
	Non-platoon speed	65 km/h
	Min Gap (r)	4m
	h	0.2s
	L_i (varying)	Table III
	k_d	2.5
	k_p Tesla	0.3
	k_p Lincoln	0.15
	k_p Rover	0.15
	C_1 Acceleration	1.0
	C_1 Deceleration	1.0
	C_2 Acceleration	1.35
	C_2 Deceleration	1.5

2) *String Stability*: String stability in platooning systems refers to the effect disturbances have on the platoon. According to [30], a platoon is said to be “string stable” if disturbances are not amplified when propagated along the platoon members. In order to study ADEPT’s string stability property, we analyze the speed and gap response of the vehicles in the platoon when disturbance is introduced. In other words, in a platoon, if vehicle i ’s deviation from the desired vehicle speed and inter-vehicle gap, i.e., the energy in vehicle i ’s error signal (e_i) is less than the energy in the error signal of vehicle- $(i - 1)$ (e_{i-1}), then the platoon is said to be string stable.

In our experiments, a minute long disturbance was introduced, where the first vehicle alternates between acceleration and deceleration every 5 seconds for a whole minute (120s - 180s) as shown in Figures 12a and 12b. After the disturbance stops, it takes a few seconds for the platoon to stabilize and we observe that vehicle speeds and inter-vehicle gaps are not at the desired levels post 180s. As previously described, the platoon is considered string stable if the energy in the error signal of a vehicle is less than that of its predecessor. If T is the set of all time steps after a vehicle experiences disturbance, then the energy of the error signal of vehicle- i is given by

$$e_i = \sum_{t=1}^n error_i \forall t \in T.$$
 In this experiment, e_i/e_{i-1} between 120s and 240s is ≈ 0.98 , which is ≤ 1 , indicating that the energy of vehicle i ’s speed and gap deviation signals is less than those of vehicle $(i - 1)$, which is a necessary and sufficient condition for a platoon to be string stable.

3) *Hyper-Parameter Tuning*: Since understanding the sensitivity of a control system to its parameters is important, in this section, we discuss the sensitivity of ADEPT system to various controller hyper-parameters. The Ploeg controller is regulated by five hyper-parameters, namely: k_p , the Gap Error Coefficient which determines how much the vehicle needs to accelerate or decelerate to maintain the desired gap with its predecessor; k_d , the Speed Error Coefficient which determines how much the vehicle needs to accelerate or decelerate to maintain the desired speed; h , the Desired Constant Time Spacing; C_1 , the Ego Controller Acceleration Coefficient which determines how much weight the controller’s

acceleration in the previous time step has on the controller’s acceleration in the current time step; and C_2 , the Predecessor Controller Acceleration Coefficient which determines how much weight the predecessor controller’s acceleration in the previous time step has on the ego controller’s acceleration in the current time step. In this section, the effects of these parameters are discussed.

Higher k_p values cause the controller to try to reach the desired gap more quickly by actively accelerating or breaking, which has the potential to create oscillation and eventually result in string instability. k_p and k_d work opposite to each other. Therefore, higher k_d values result in the vehicle trying to reach the desired gap slower, which decreases the chances of oscillations instability.

Increasing the desired time spacing, h , allows vehicles more time to react to disturbances. However, as h increases causing vehicles to react slower to disturbances, we notice an increase in the deviation from the desired gap. In other words, high h values result in high inter-vehicle gaps potentially causing string instability, while low h values may cause vehicles to crash.

Lastly, C_1 and C_2 determine how much weight the predecessor and ego vehicles’ acceleration at time $(t - 1)$ should have on determining the ego vehicle’s acceleration at time t . Higher C_1 values will lower the weight of the predecessor’s acceleration in the calculation of the ego vehicle’s acceleration and therefore the ego vehicle’s deviation from the desired gap tends to increase. On the other hand, higher C_2 values will make the ego vehicle align more with the predecessor and the deviation from the desired gap decreases. However, it may lead to string instability.

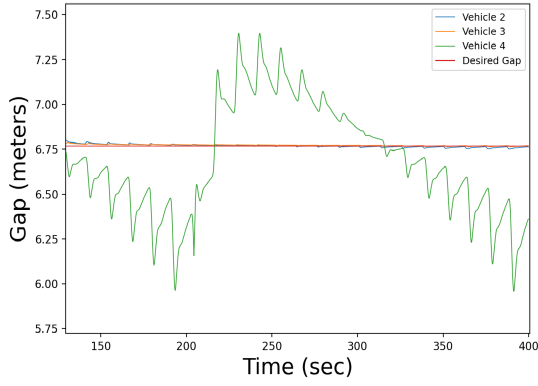
In general, what we observe in our experiments is that too low or too high value of any hyper-parameter lead to oscillations and string instability. In our current implementation of ADEPT, the controller hyper-parameters are tuned empirically in order to guarantee string stability. As future work, we plan to explore machine learning techniques to automatically adjust the controller’s hyper-parameters.

D. Platooning on Curved Roads

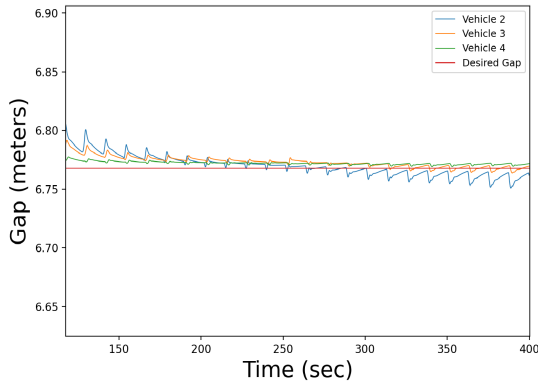
In this section we evaluate ADEPT’s performance when the platoon is driving on curved roads employing the techniques described in Section III-B.

1) *Road Curvature Calculation Performance*: As described in Section III-B2, accurately determining the radius of curvature of the road dynamically is critical in order to maintain the desired inter-vehicle gap.

Every vehicle in ADEPT needs to determine the distance to its predecessor to ensure that the inter-vehicle gap stays as close to the desired gap as possible. The desired gap in our experiments is 6.77m (see Section V-C) and, we stay within 99% of the desired gap, i.e., in ADEPT all vehicles but for the last maintains a maximum deviation of 0.07m from the desired gap. In curved sections of the road, the distance (l) to a vehicle’s predecessor is calculated as $l = r\theta$, where r is the radius of curvature of the road and θ is the angle subtended by the arc between the two vehicles. The road’s radius of curvature is calculated based on the vehicle’s own position



(a) Last vehicle using avg of Euclidean & Manhattan distance



(b) Last vehicle use the radius calculated by its predecessor

Fig. 13. Gap approximation for the last vehicle.

as well as its predecessor's and successor's positions. Note that the first vehicle does not need to determine its distance with its predecessor as it does not have a predecessor. The last vehicle has to approximate the road's radius of curvature since it does not have a successor. We evaluated two approximation methods for the last vehicle to calculate the road's radius of curvature. In the first method, the distance between the last vehicle at position (x_3, y_3) and its predecessor at position (x_2, y_2) was approximated as the average of the Euclidean and the Manhattan distances between them. As shown in Figure 13a, we observe relatively high gap errors for the last vehicle (v_4) up to 0.63m, a 11.9% deviation from the desired gap. To address this issue, the second approximation method to calculate the road's radius of curvature for the last vehicle of the platoon uses the radius calculated by its predecessor. This information is propagated as part of pheromone-messages. This approach reduced the deviation from the desired gap substantially to a maximum of 0.05m, i.e., 0.8% deviation as shown in Figure 13b and on par with other vehicles in the platoon. One interesting observation is that, when the road is curved, the inter-vehicle gap gradually decreases. In our experiments, the gap decreases at a rate of 10^{-5} meters per minute as shown in Figure 13b. For a vehicle traveling at 50kmph, it is 10^{-5} meters for every 833 meters. In real world scenarios, only portions of roads are curved so that when the platoon is on the road's straight portions, the inter-vehicle gap stabilizes to the desired gap as seen in Figure 12b.

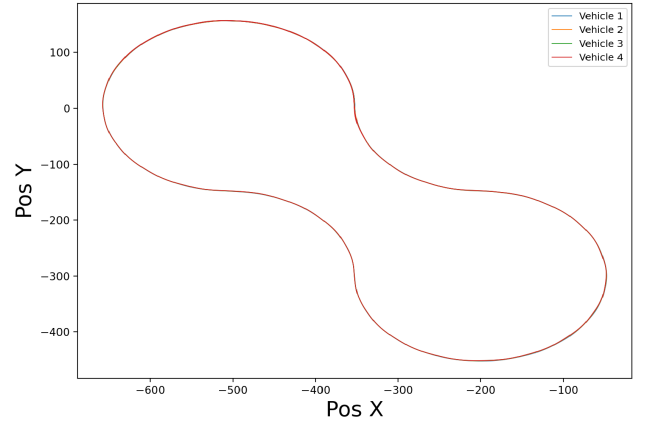


Fig. 14. Dumbbell-shaped trajectory used to evaluate ADEPT's performance in curved roads.

TABLE II

PEARSON CORRELATION COEFFICIENT MEASURING SIMILARITY OF TRAJECTORIES FOLLOWED BY PLATOON VEHICLES DRIVING ON A DUMBELL SHAPED CIRCUIT

VID	1	2	3	4
1	$x: 1.00$ $y: 1.00$	$x: 0.999$ $y: 0.999$	$x: 0.996$ $y: 0.996$	$x: 0.991$ $y: 0.991$
2	$x: 0.999$ $y: 0.999$	-	-	-
3	-	$x: 0.999$ $y: 0.999$	-	-
4	-	-	$x: 0.999$ $y: 0.999$	-

2) *Vehicle Following Performance*: In a platoon, for the ego vehicle to follow the trajectory of its predecessor when driving on curved roads, it must determine the required steering angle dynamically. As described in Section III-B4, ADEPT uses the angle of predecessor vehicle extracted from the vehicle's camera image. The inspiration for using the predecessor's image came from the way vehicles align themselves to stay within a lane using the lane-assist feature. For obstacle avoidance, we use a mechanism similar to gap maintenance using LiDAR image as explained in Section III-B5.

In the current ADEPT implementation, vehicle i follows vehicle $(i - 1)$ by using the color of the vehicle and comparing it against the camera image that is constantly fed to the platooning layer. The image is processed to calculate the steering angle as explained in Section III-B4.

We test the efficacy of this method by simulating a curved road with radius of curvature equal to 152.4m. According to Chapter-200 of the Highway Design Manual published by the California Department of Transportation in 2022 [43], the curvature radius of horizontal curves for a maximum speed of 35mph at 0ft elevation is equal to 500ft. This corresponds to 152.4m for 50kmph. In these experiments, we used a dumbbell shaped road that includes both left and right turns as illustrated in Figure 14. Note that trajectories followed by each of the four simulated vehicles are very similar to one another and thus they are seen as one solid line in Figure 14.

To measure the accuracy of ADEPT's vehicle following technique, the (x, y) coordinates of each vehicle in the platoon are considered as individual signals. The path taken by each vehicle is compared against that of the lead vehicle which represents the ground truth. Additionally, the path taken by

TABLE III

WEBOTS VEHICLE MODEL CHARACTERISTICS TM3: TESLA
MODEL3, LMKZ: LINCOLN MKZ, RR: RANGE ROVER

Veh	Mass (Kg)	Acc (m/s ²)	Max Torque (Nm)	Max Power (Watt)	Wheel Base (m)	Wheel Damping Const
TM3	1847	6.94	639	147000	2.94	3
LMKZ	1950	5.58	-	-	2.94	2
RR	2700	5.91	-	-	2.71	5

vehicle v_i is compared against that of vehicle v_{i-1} , which the vehicle v_i follows. We use the Pearson correlation coefficient, a popular signal processing technique used to compare signals, to compare the trajectories taken by the vehicles. The Pearson correlation coefficient is defined as the ratio between the covariance of two vectors (x, y) and the product of their standard deviations. Therefore, it is a normalized measurement of the covariance, such that the result is always between $(-1, 1)$. The closer the Pearson coefficient is to 1, the more similar the two signals (here trajectories) are. Signals with Pearson coefficient values greater than 0.75 are generally considered to be strongly correlated. Table II summarizes the Pearson correlation coefficients of the platoon. We observe that trajectories taken by successive vehicles yield Pearson coefficients very close to 1, e.g., 0.999. While the farther the vehicle is from the lead vehicle, the lesser its path is correlated with the first vehicle's trajectory, they are still very strongly correlated with Pearson coefficients greater than 0.99. This confirms that ADEPT's vehicle following method is quite effective.

E. ADEPT's Maneuver Performance

This section focuses on evaluating ADEPT's emergent-behavior based maneuvers. More specifically, we measure the time taken to perform the Join and Exit maneuvers. Implementation and performance evaluation of ADEPT's Merge maneuver is one of our future work goals. Rather than compare ADEPT's maneuvers to other platooning approaches, e.g., centralized platooning, as we did in our previous study [7], the goal of our analysis here is to evaluate ADEPT's emergent based platooning when the physics of the underlying vehicles is considered.

For the maneuver performance experiments, we simulated three vehicles using three different vehicle models available in the Webots simulator, viz., Tesla Model3, Lincoln MKZ and Range Rover. The vehicle models used are part of Webots' standard library, have different characteristics, and are calibrated based on their actual physical and engine features. Additionally, as previously noted, we assume that each vehicle is equipped with a GPS sensor to determine its position, a LiDAR camera for obstacle detection and a video camera for car following. Table III summarizes the different vehicle characteristics, where WheelBase is the distance from the front wheel axel to the rear wheel axel and WheelDampingConstant is used to simulate the effect of friction. MaxTorque ($maxT$) and MaxPower ($maxP$) are used to calculate the output torque of electric vehicles using the equation -

$$\text{Output Torque} = \min(maxT, maxP \times 60 / (2\pi \text{ rpm})) \quad (7)$$

TABLE IV

JOIN MANEUVER EXPERIMENT SUMMARY: TIME (IN SECONDS) TAKEN TO
REACH DIFFERENT STAGES FOR THE DIFFERENT VEHICLE MODELS
TM3: TESLA MODEL3, LMKZ: LINCOLN MKZ, RR: RANGE ROVER

Veh	Start	Start Join	Joined	Platoon Speed	Desired Gap	Total Time
TM3	8.21	43.41	44.71	57.41	70.21	29.8
LMKZ	8.21	42.11	43.41	69.41	131.61	89.1
RR	8.21	42.51	43.81	70.01	123.01	80.5

For non-electric vehicles, the output torque for a given rpm is calculated using the equation -

$$\text{Output Torque} = f.rpm^2 + e.rpm + d \quad (8)$$

where (d, e, f) are engine function coefficients that are used to determine the torque of the combustion engine and is equal to $(150, 0.1, 0)$ respectively, which are the defaults in Webots.

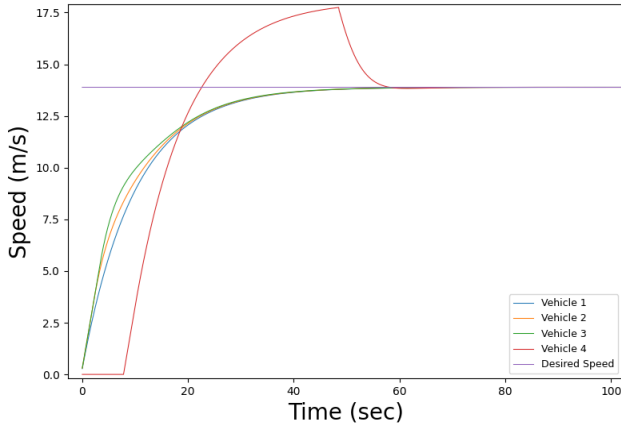
Due to space limitations, we present only graphs for Tesla Model3 here as other vehicles follow the same pattern. Results for all vehicle models are summarized in Tables IV and V for join and exit maneuvers, respectively. Videos of the different maneuvers we simulated are published in public domain [44].

In the remainder of this section, vehicles $\{1, 2, 3, 4\}$ are referred to as $\{v_1, v_2, v_3, v_4\}$, with v_1 being the first vehicle and v_4 being the last vehicle in the platoon.

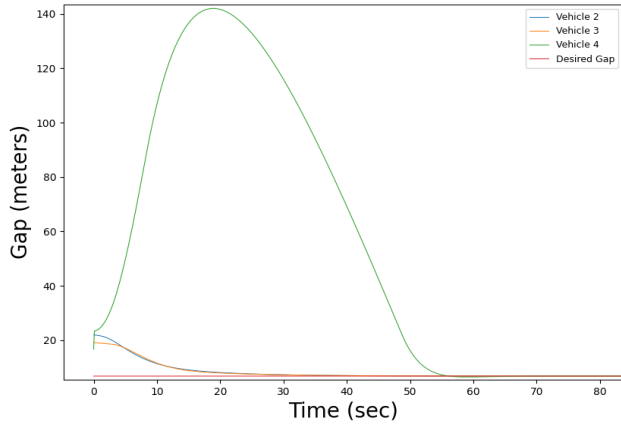
1) *Join Maneuver*: To evaluate the performance of ADEPT's Join, we conducted the following experiment: initially, all vehicles are parked equidistant from each other. The first three vehicles are part of pre-formed platoon while vehicle v_4 is not part of any platoon. As illustrated in Figure 15a, initially, all vehicles except v_4 enter the system at time 0s; v_4 enters the system at 8.21s. The max speed of non-platooned vehicles is set to be 60kmph, 10kmph higher than the platoon speed of 50kmph. The speed of non-platooned vehicles is set higher to enable them to catch-up with a platoon in order to join. Since v_4 enters the system later, the initial gap with its predecessor (v_3) is high as shown in Figure 15b. Both v_2 and v_3 have vehicles in front of them and therefore have to adjust their speed by accelerating and decelerating. Vehicles v_1 and v_4 don't have any vehicles in front (initially for v_4) and therefore, have initial spike in acceleration but smoothly decelerate to the desired speed.

When v_4 gets in wireless communication range of v_3 , it starts receiving v_3 's pheromone-messages including v_3 's ID. When v_4 gets within platoon range, which happens at 30m in our experiments, a configurable parameter, it starts including v_3 's ID in its own pheromone-messages, which happens at 43.41s.

When v_4 receives a minimum of n pheromone-messages from v_3 (in our current implementation n is set to 10), it accepts v_3 's platoon ID and becomes part of the platoon, which happens at 44.71s. From the time v_4 becomes part of the platoon, it starts to decelerate since its current speed is higher compared to the platoon speed. This corresponds to the smooth decrease in v_4 's speed shown in Figure 15a. While v_4 exhibited somewhat jerky deceleration between



(a) Speed response during join maneuver



(b) Gap response during join maneuver

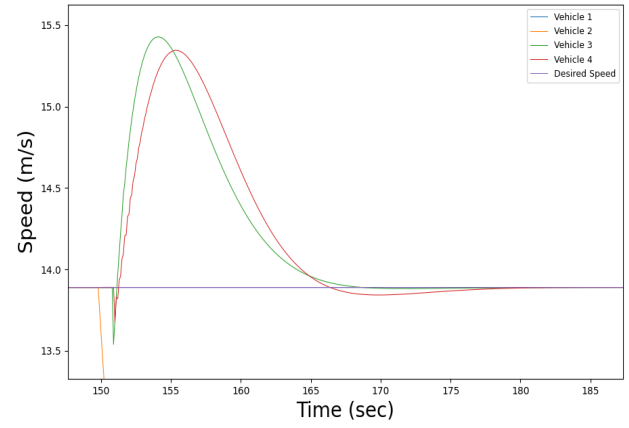
Fig. 15. Speed & gap response during Join maneuver for Tesla Model3.

(0, -1.2) m/s^2 when trying to adjust to the platoon speed,³ the inter-vehicle gap between v_3 and v_4 gradually decreases and eventually stabilizes to the desired gap as shown in Figure 15b. Table IV summarizes the Join maneuver results for the three different vehicle types we used in our experiments. As shown in Table IV, when v_4 joins the platoon, it slowly decelerates and reaches the desired platoon speed at 57.41s and the desired gap at 70.21s.

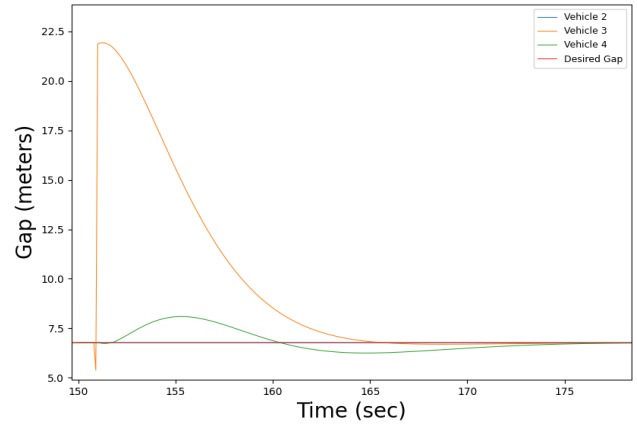
2) *Exit Maneuver*: Once v_4 becomes part of the platoon, we start the Exit maneuver. This happens at 150.11s which allows enough time for the vehicles to reach steady state, driving at the target platoon speed. For the Exit maneuver experiment, we chose v_2 to exit the platoon and, once it exits, it changes its controller to CC. The target speed after the maneuver is complete is set to zero. During the maneuver, v_2 's speed gradually drops to zero. Therefore, v_3 's gap with its new predecessor v_1 shoots up as seen in Figure 16b and to close the gap, v_3 accelerates as illustrated in Figure 16a.

When v_2 exits the platoon, v_3 accelerates to close the gap. Similar to disturbance events, we need to guarantee that trying to close the gap due to the exit of a vehicle does not cause string instability, i.e., the disturbance does not

³This is not shown in Figure 15a due to space limitation. Additionally, the goal here is to demonstrate platoon by focusing on platoon speed and gap.



(a) Speed response during exit maneuver



(b) Gap response during exit maneuver

Fig. 16. Speed & gap response during Exit maneuver for Tesla Model3.

TABLE V

EXIT MANEUVER EXPERIMENT SUMMARY: TIME (IN SECONDS) TAKEN TO REACH DIFFERENT STAGES FOR THE DIFFERENT VEHICLE MODELS
TM3: TESLA MODEL3, LMKZ: LINCOLN MKZ, RR: RANGE ROVER

Veh	Start Exit	Notice Exit	Platoon Speed	Desired Gap	Total Time
TM3	150.11	151.11	166.11	179.61	28.5
LMKZ	150.11	151.11	220.21	240.11	89.0
RR	150.11	151.11	173.81	239.51	88.4

propagate towards the tail of the platoon. Note that during the Exit maneuver (16a), v_4 's speed is less than v_3 's which demonstrates string stability. The maximum acceleration and deceleration of the controller are set to 1.5 and $-3.5m/s^2$, respectively. Therefore, vehicle acceleration and deceleration do not go beyond that. These acceleration/deceleration values are recommended by several studies [45], [46] as "comfortable acceleration" for autonomous passenger vehicles. They point out that if acceleration limits are increased then more jerky motion will be experienced which is not desirable and if the limits are decreased, it takes much longer to attain platoon speed. Table V summarizes the Exit maneuver results for the three different vehicle types we used in our experiments.

One result that stands out in Tables IV and V is Tesla Model3's low total maneuver time compared to the other vehicle models. In order to investigate why this is the case, we plot the speed of the three vehicles during the Join and

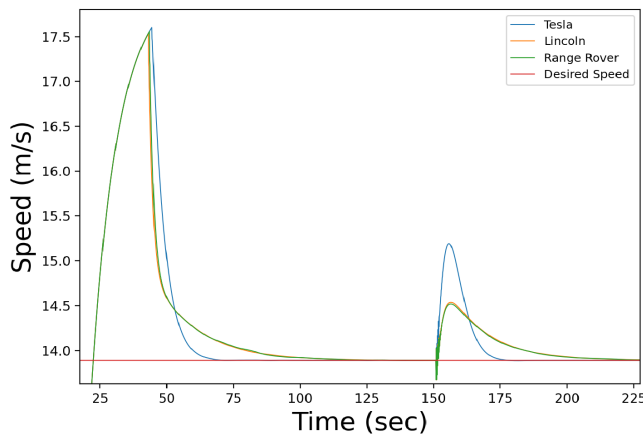


Fig. 17. Speeds of different cars during join and exit maneuver.

Exit maneuvers in Figure 17. We observe that Tesla Model3 has the ability to increase and decrease its speed much faster which is why it is able to achieve the desired gap and speed faster than the other models.

VI. CONCLUSION

Platooning is a technique to improve fuel efficiency, road capacity, driver and road safety. We propose ADEPT - the Adaptive Decentralized Emergent Behavior based Platooning System for Autonomous Vehicles, whose emergent behavior draws inspiration from biological systems. In ADEPT, platooning vehicles follow a set of simple emergent rules, from which platooning maneuvers emerge. Our current ADEPT implementation includes three main platooning maneuvers, viz., Join, Exit and Merge as they are sufficient for a complete freeway platooning system. Because of its decentralized nature, ADEPT does not suffer from the drawbacks of centralized platooning such as single point of failure, maneuver serialization and limited platoon length. We also developed an emergent rule set sufficient to implement the three main freeway platooning maneuvers mentioned above. In this paper, we evaluated the interplay between ADEPT and the underlying wireless communication infrastructure. We showed that ADEPT is able to better utilize the network compared to the centralized platooning approach. Additionally, we also implemented ADEPT's controller and platooning algorithms on a physics simulator to understand and evaluate the interaction between ADEPT's emergent platooning and the physics of the underlying vehicle. Our experiments and results demonstrated that ADEPT's controller is string stable when subject to different maneuvers and disturbances. In addition to evaluating ADEPT's controller stability, we also tested ADEPT's behavior when driving in curved roads. Our experimental results show that our dynamic radius of curvature determination method helps maintain inter-vehicle gap at 99% of the desired gap, including on curved roads. Inspired by the way vehicles align themselves to stay within a lane using the lane-assist feature, we used camera images of the predecessor vehicle to implement vehicle following and maintain platoon trajectory. We showed that this mechanism provides high accuracy. Videos of the different maneuvers we simulated are published in public domain [44].

In future work, we plan to explore machine learning techniques for hyper-parameter tuning as it will also allow us to adjust the hyper-parameters dynamically. We will also consider additional real world scenarios that include larger number of platooning vehicles as well as various weather and road conditions. Additionally, we will implement and evaluate ADEPT's Merge maneuver using the Webots physics simulator [39]. Considering the challenges of urban platooning such as coexisting with traditional non-autonomous vehicles, handling intersections, signal lights, cyclists, pedestrians, etc., are also planned for future work.

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